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delay(1000);
/* Loads the default angles for each
 * joint */
Load_Default();
/* Start polling the switches */
manual();

/* Stops controller execution */
while(1);
}

/*-----Delay-----*/
void delay(unsigned int count)
{
    unsigned int i;
    for(i=0;i<count;i++);
}

/*-----Loads the base servo-----*/
void Load_Servo()
{
    CCPRL = reg_CCP[deg_servo];
    CCPCON = reg_CON[deg_servo & 0x03];
}

/*---Loads the shoulder servo---*/
void Load_Base()
{
    CCPRL = reg_CCP[deg_base];
    CCPCON = reg_CON[deg_base & 0x03];
}

/*-----Loads the elbow servo-----*/
void Load_Elbow()
{
    PDCR = reg_PDC[deg_elbow >> 4];
    PDCR = reg_PDC[deg_elbow & 0x0F];
}

/*-----Loads the wrist servo-----*/
void Load_Wrist()
{
    PDCR = reg_PDC[deg_wrist >> 4];
    PDCR = reg_PDC[deg_wrist & 0x0F];
}

/*---Loads the rotating grip servo---*/
void Load_R_Grip()
{
    PDCR = reg_PDC[deg_r_grip >> 4];
    PDCR = reg_PDC[deg_r_grip & 0x0F];
}

/*-----Loads the grip servo-----*/
void Load_Grip()
{
    PDCR = reg_PDC[deg_grip >> 4];
    PDCR = reg_PDC[deg_grip & 0x0F];
}

/* Loads the default servo joint
angles */
void Load_Default()
{
    while(deg_base != 15)
    {
        if (deg_base < 15)
        {
            deg_base++;
            Load_Base();
            delay(500);
        }
        else
        {
            deg_base--;
            Load_Base();
            delay(500);
        }
    }
    Load_Base();
    while(deg_elbow != 5)
    {
        if (deg_elbow < 5)
        {
            deg_elbow++;
            Load_Elbow();
            delay(300);
        }
        else
        {
            deg_elbow--;
            Load_Elbow();
            delay(300);
        }
    }
    Load_Elbow();
    delay(1000);
    while(deg_servo != 31)
    {
        if (deg_servo < 31)
        {
            deg_servo++;
            Load_Servo();
            delay(300);
        }
        else
        {
            deg_servo--;
            Load_Servo();
            delay(300);
        }
    }
    Load_Servo();
    while(deg_wrist != 10)
    {
        if (deg_wrist < 10)
        {
            deg_wrist++;
            Load_Wrist();
            delay(300);
        }
        else
        {
            deg_wrist--;
            Load_Wrist();
            delay(300);
        }
    }
    Load_Wrist();
    while(deg_r_grip != 31)
    {
        if (deg_r_grip < 31)
        {
            deg_r_grip++;
            Load_R_Grip();
            delay(300);
        }
        else
        {
            deg_r_grip--;
            Load_R_Grip();
            delay(300);
        }
    }
    Load_R_Grip();
}

/* Polls switches and loads values
for manual control */
void Manual()
{
    while(1)
    {
        /*-----Base servo-----*/
        if((RD5 == 1)&&(deg_servo < 63))
        {
            deg_servo++;
            Load_Servo();
            delay(300);
        }
        else if((RD4 == 1)&&(deg_servo > 0))
        {
            deg_servo--;
            Load_Servo();
            delay(300);
        }
        /*-----Shoulder servo-----*/
        if((RD7 == 1)&&(deg_base < 63))
        {
            deg_base++;
            Load_Base();
            delay(500);
        }
        else if((RD6 == 1)&&(deg_base > 13))
        {
            deg_base--;
            Load_Base();
            delay(500);
        }
        /*-----Elbow servo-----*/
        if((RD5 == 1)&&(deg_elbow < 63))
        {
            deg_elbow++;
            Load_Elbow();
            delay(500);
        }
        else if((RD4 == 1)&&(deg_elbow > 0))
        {
            deg_elbow--;
            Load_Elbow();
            delay(500);
        }
        /*-----Wrist servo-----*/
        if((RD3 == 1)&&(deg_wrist < 63))
        {
            deg_wrist++;
            Load_Wrist();
            delay(300);
        }
        else if((RD2 == 1)&&(deg_wrist > 0))
        {
            deg_wrist--;
            Load_Wrist();
            delay(300);
        }
        /*---Rotate Grip servo---*/
        if ((RD6 == 1)&&(deg_r_grip < 63))
        {
            deg_r_grip++;
            Load_R_Grip();
            delay(500);
        }
        else if((RD7 == 1)&&(deg_r_grip > 0))
        {
            deg_r_grip--;
            Load_R_Grip();
            delay(500);
        }
        /*-----Grip servo-----*/
        if((RD1 == 1)&&(deg_grip < 55))
        {
            deg_grip++;
            Load_Grip();
            delay(300);
        }
        else if((RD0 == 1)&&(deg_grip > 15))
        {
            deg_grip--;
            Load_Grip();
            delay(300);
        }
    }
}

```